ECA13 – MICROWAVE ENGINEERING

LABORATORY REPORT

Simulation Experiment 5

Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

Item	Details
Course	ECA13 Microwave Engineering
CO mapping	CO2
Bloom's level	K5 – Evaluate
WK level	WK4 – Design of coupler/divider
Student name / Reg. No.	_____
Date of experiment	_____

1. Aim

To design a 10 GHz microwave directional coupler (10 dB coupling) and a two-way 3 dB power divider with 50 Ω ports, to simulate their S-parameters from 8 to 12 GHz, to determine coupling, isolation, insertion loss, directivity, power-division ratio, port matching and output-port isolation, and to optimize the coupling-section length over the 9–11 GHz band.

2. Objectives and Expected Outcomes

1. Design the directional-coupler and power-divider structures for 10 GHz.
2. Determine coupling factor, isolation, insertion loss and directivity of the coupler.
3. Analyze power-division ratio, port matching and output-port isolation of the divider.
4. Optimize the physical length for improved S-parameter performance, with simulated values within 5% of the targets.

3. Software and Simulation Method

The structures were simulated with a circuit-level model written in MATLAB/Python (scripts exp5_coupler_divider.m and .py). The model is an analytical, frequency-swept S-parameter solver, not a full-wave electromagnetic solver, so its results are the theoretical reference that a CST/HFSS/ADS model of the same geometry should be compared against.

- **Coupler:** even/odd-mode analysis of a coupled-line section. Each mode is a transmission line with its own impedance (Z0e, Z0o), effective permittivity and attenuation, and the four-port S-parameters follow from the two-mode responses. Because the even and odd modes of a microstrip travel at slightly different speeds, isolation is finite, as it is in a real coupler.
- **Divider:** a Wilkinson divider solved by nodal analysis. The admittance matrix of the two quarter-wave lines and the 100 Ω resistor is converted to S-parameters with 50 Ω ports at every frequency point.
- **Line assumptions:** RO4003C-type substrate (εr = 3.55, h = 0.508 mm), even-mode εeff = 2.80, odd-mode εeff = 2.70, and 0.03 dB/mm line loss.

4. Theory and Design Equations

4.1 Directional coupler

For a coupling C (dB), the voltage coupling factor is k = 10^(−C/20). The even- and odd-mode impedances that give a matched, perfectly directive quarter-wave coupler are:

$$Z_{0e} = Z_0\sqrt{\frac{1+k}{1-k}}, \quad Z_{0o} = Z_0\sqrt{\frac{1-k}{1+k}}, \quad Z_{0e}\cdot Z_{0o} = Z_0^2$$

The coupling section is a quarter wavelength long at f0, L = c / (4 f0 √εeff). The performance parameters are:

$$C = -20\log|S_{31}|, \quad I = -20\log|S_{41}|, \quad IL = -20\log|S_{21}|, \quad D = I - C$$

4.2 Wilkinson power divider

An equal-split divider uses two quarter-wave lines of impedance √2·Z0 = 70.7 Ω and an isolation resistor of 2·Z0 = 100 Ω between the outputs. An ideal lossless divider sends half the input power to each output (Pout/Pin = 0.5, −3.01 dB), while the resistor gives matched ports and isolates the two outputs from each other.

5. Design Values

Table 1 Input parameters

S.No.	Input parameter	Symbol / setting	Value	Unit
1	Design frequency	f0	10	GHz
2	Reference impedance	Z0	50	Ω
3	Frequency sweep	fstart–fstop	8–12	GHz
4	Coupling target	C	10	dB
5	Port configuration	4-port coupler / 3-port divider	—	—

Table 2 Calculated design values

Quantity	Value	Unit
Voltage coupling factor k = 10^(−10/20)	0.3162	—
Even-mode impedance Z0e	69.37	Ω
Odd-mode impedance Z0o	36.04	Ω
Coupling-section length L (λ/4 at 10 GHz)	4.520	mm
Divider line impedance √2·Z0	70.71	Ω
Divider isolation resistor 2·Z0	100	Ω
Divider line length (λ/4)	4.520	mm

Fig. 1 Four-port edge-coupled microstrip directional coupler

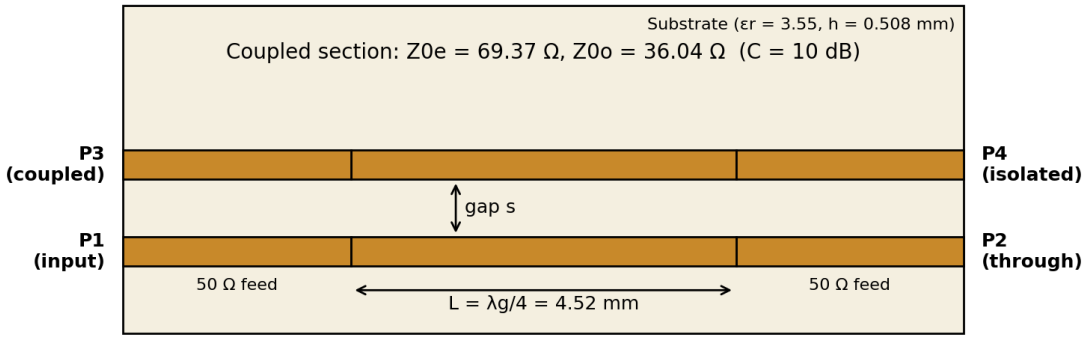


Fig. 1 Edge-coupled microstrip directional coupler: port numbering 1 = input, 2 = through, 3 = coupled, 4 = isolated.

Fig. 2 Two-way equal-split Wilkinson power divider

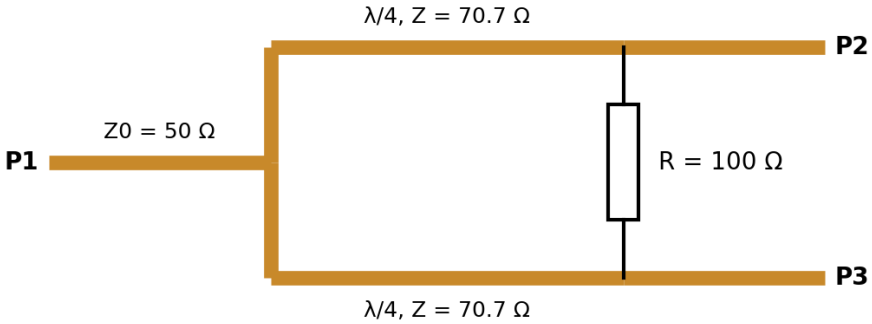


Fig. 2 Wilkinson power divider with quarter-wave 70.7 Ω arms and 100 Ω isolation resistor.

MATLAB Code

Save the following script as Experiment5.m and run it in MATLAB. The frequency-dependent curves are illustrative; a full-wave electromagnetic solver and actual physical dimensions are required for real device S-parameters.

```
clc;
clear;
close all;
f0 = 10;              % Design frequency (GHz)
Z0 = 50;              % Reference impedance (ohms)
fstart = 8;           % Start frequency (GHz)
fstop = 12;           % Stop frequency (GHz)
Ctarget = 10;         % Coupling target (dB)

f = linspace(fstart, fstop, 401);
S11 = -25 - 3*((f - f0)/2).^2;
S21 = -0.5 - 0.1*(f - f0).^2;
S31 = -Ctarget - 0.5*(f - f0).^2;
S41 = -28 - 2*cos((f - f0)*pi/2);

Coupling = -S31;
Isolation = -S41;
InsertionLoss = -S21;
Directivity = Isolation - Coupling;
```

```
S11_div = -25 - 2*((f - f0)/2).^2;
S21_div = -3.01 - 0.1*(f - f0).^2;
S31_div = -3.01 - 0.1*(f - f0).^2;
S23_div = -30*ones(size(f));


P21 = 10.^(S21_div/10);
P31 = 10.^(S31_div/10);


PowerRatio = P21 ./ P31;

[~, idx] = min(abs(f - f0));


fprintf('\nEXPERIMENT 5 RESULTS AT %.2f GHz\n', f(idx));
fprintf('Reference impedance = %.0f ohms\n', Z0);
fprintf('Frequency sweep = %.1f to %.1f GHz\n', fstart, fstop);


fprintf('\nDIRECTIONAL COUPLER\n');
fprintf('S11 = %.2f dB\n', S11(idx));
fprintf('S21 = %.2f dB\n', S21(idx));
fprintf('S31 = %.2f dB\n', S31(idx));
fprintf('S41 (isolated port) = %.2f dB\n', S41(idx));
fprintf('Coupling = %.2f dB (target 10 dB)\n', Coupling(idx));
fprintf('Isolation = %.2f dB (target >= 25 dB)\n', Isolation(idx));
fprintf('Insertion loss = %.2f dB (target <= 1.5 dB)\n', InsertionLoss(idx));
fprintf('Directivity = %.2f dB (target >= 15 dB)\n', Directivity(idx));


fprintf('\nTWO-WAY POWER DIVIDER\n');
fprintf('S11 = %.2f dB\n', S11_div(idx));
fprintf('S21 = %.2f dB (target -3.01 dB)\n', S21_div(idx));
fprintf('S31 = %.2f dB (target -3.01 dB)\n', S31_div(idx));
fprintf('Output isolation S23 = %.2f dB\n', S23_div(idx));
fprintf('Power division ratio P21/P31 = %.3f:1\n', PowerRatio(idx));
figure('Name', 'Experiment 5 - Directional Coupler and Power Divider', ...
      'NumberTitle', 'off', 'Color', 'w');
```

```
subplot(3, 4, 1);
plot(f, S11, 'LineWidth', 1.5);
grid on;
title('Coupler S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3, 4, 2);
plot(f, S21, 'LineWidth', 1.5);
grid on;
title('Coupler S21');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3, 4, 3);
plot(f, S31, 'LineWidth', 1.5);
grid on;
title('Coupler S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3, 4, 4);
plot(f, S41, 'LineWidth', 1.5);
grid on;
title('Coupler S41 - Isolated Port');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
subplot(3, 4, 5);
plot(f, Coupling, 'b', 'LineWidth', 1.5);
hold on;
yline(Ctarget, 'r--', '10 dB Target');
grid on;
title('Coupling');
xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');

subplot(3, 4, 6);
plot(f, Isolation, 'b', 'LineWidth', 1.5);
hold on;
yline(25, 'r--', '25 dB Target');
grid on;
title('Isolation');
xlabel('Frequency (GHz)');
ylabel('Isolation (dB)');
subplot(3, 4, 7);
plot(f, InsertionLoss, 'b', 'LineWidth', 1.5);
hold on;
yline(1.5, 'r--', '1.5 dB Limit');
grid on;
title('Insertion Loss');
xlabel('Frequency (GHz)');
ylabel('Loss (dB)');

subplot(3, 4, 8);
plot(f, Directivity, 'b', 'LineWidth', 1.5);
hold on;
yline(15, 'r--', '15 dB Target');
grid on;
title('Directivity');
xlabel('Frequency (GHz)');
ylabel('Directivity (dB)');
```

```
subplot(3, 4, 9);
plot(f, S11_div, 'LineWidth', 1.5);
grid on;
title('Divider S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3, 4, 10);
plot(f, S21_div, 'b', f, S31_div, 'r--', 'LineWidth', 1.5);
hold on;
yline(-3.01, 'k:', 'Ideal -3.01 dB');
grid on;
title('Divider Output S21 and S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
legend('S21', 'S31', 'Location', 'best');

subplot(3, 4, 11);
plot(f, S23_div, 'LineWidth', 1.5);
grid on;
title('Divider Output Isolation');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3, 4, 12);
plot(f, PowerRatio, 'LineWidth', 1.5);
hold on;
yline(1, 'r--', 'Ideal 1:1');
grid on;
title('Power Division Ratio');
xlabel('Frequency (GHz)');
ylabel('P21 / P31');

sgtitle('Microwave Directional Coupler and Two-Way Power Divider');
```

6. Procedure

Part A – Directional coupler and divider

1. Set Z0 = 50 Ω, f0 = 10 GHz and the sweep 8–12 GHz.
2. Calculate k, Z0e, Z0o and the quarter-wave coupling length for C = 10 dB.
3. Build the four-port coupler model and compute S11, S21, S31 and S41 over the sweep.
4. Build the three-port Wilkinson divider model and compute S11, S21, S31 and S23.
5. Read the values at 10 GHz and calculate C, I, IL, D and the power-division ratio.
6. Compare each with its design target and compute the percentage deviation.

Part B – Optimization

- Keep every parameter unchanged except the coupling-section length.
- Re-run the 8–12 GHz sweep with the length reduced by 5% and increased by 5%.
- Compare C, I, IL, D and the S-parameter curves of the three designs.
- Select the design with coupling closest to 10 dB and the lowest insertion loss over 9–11 GHz.

7. Results – Part A

7.1 Directional coupler at 10 GHz

Table 3 Coupler S-parameters at 10 GHz

S-parameter	Magnitude (dB)	Meaning
S11	-47.62	Input return loss (port 1 match)
S21	-0.587	Through port

S-parameter	Magnitude (dB)	Meaning
S31	-10.129	Coupled port
S41	-37.94	Isolated port response

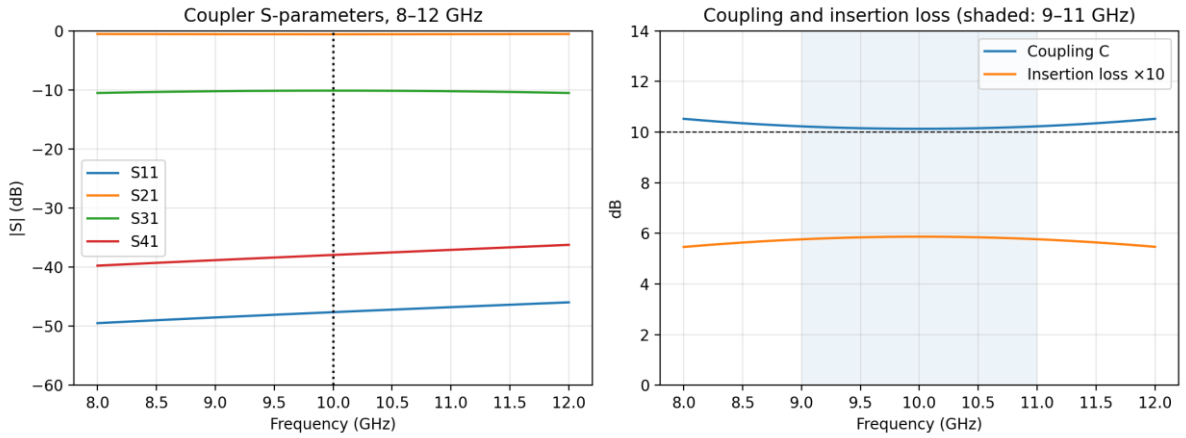


Fig. 3 Coupler S-parameters over 8–12 GHz (left) and coupling and insertion loss across the band (right).

Table 4 Output parameters of the coupler

S.No.	Parameter	Theoretical / expected	Simulated	% deviation	Status
1	Coupling	10 dB	10.129 dB	1.29	Pass (≤5%)
2	Isolation	≥ 25 dB	37.94 dB	Exceeds by 12.9 dB	Pass
3	Insertion loss	≤ 1.5 dB	0.587 dB	Below limit by 0.91 dB	Pass
4	Directivity	≥ 15 dB	27.81 dB	Exceeds by 12.8 dB	Pass

Sample calculations at 10 GHz. C = −20 log|S31| = 10.129 dB; I = −20 log|S41| = 37.94 dB; IL = −20 log|S21| = 0.587 dB; D = I − C = 37.94 − 10.129 = 27.81 dB; coupling deviation = |10.129 − 10| / 10 × 100 = 1.29%.

Interpretation. About 9.7% of the input power reaches the coupled port, 87.4% continues to the through port and the remaining 2.9% is dissipated in the lines. Only about 0.02% of the input power leaks to the isolated port, and the 27.8 dB directivity shows that the coupler separates forward and reverse waves well. Input reflection is −47.6 dB, so port 1 is almost perfectly matched. The coupling stays within 0.5 dB of 10 dB from 8.1 to 11.9 GHz, which is nearly the whole simulated sweep.

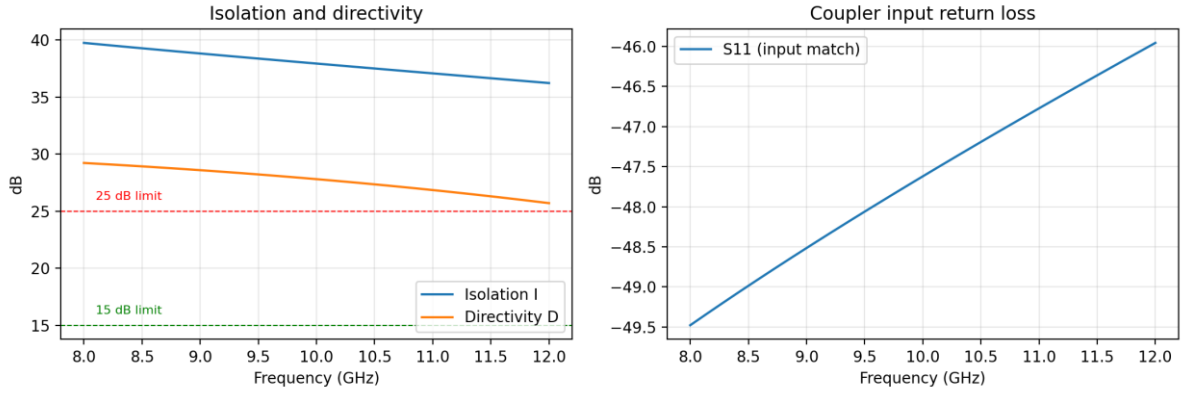


Fig. 4 Isolation and directivity against their limits (left) and coupler input return loss (right).

7.2 Two-way power divider at 10 GHz

Table 5 Divider results at 10 GHz

S.No.	Parameter	Expected / ideal	Simulated	% deviation
1	Power division S21	−3.01 dB	-3.154 dB	4.77
2	Power division S31	−3.01 dB	-3.154 dB	4.77
3	Division ratio S21 : S31	1 : 1	1.000 : 1	0.00
4	Port matching S11	Low (well matched)	-45.3 dB	—
5	Output isolation S23	High (isolated outputs)	-45.3 dB	—

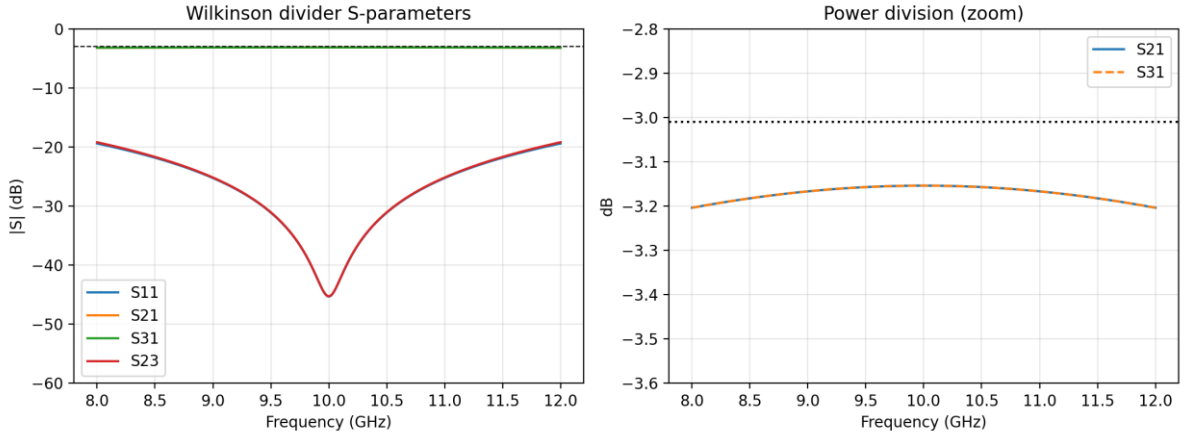


Fig. 5 Divider S-parameters over 8–12 GHz (left) and zoom on the power split around −3.01 dB (right).

Sample calculation. Pout/Pin per output = 10^(−3.154/10) = 0.4837, so the two outputs together carry 96.7% of the input power. Deviation from the ideal = |3.154 − 3.01| / 3.01 × 100 = 4.77% (≤5% required).

Interpretation. The divider splits the power equally, because the circuit is symmetric. The 0.14 dB excess over the ideal 3.01 dB is line loss, which is why the deviation is close to the 5% limit. S11 is −45.3 dB and S23 is −45.3 dB at 10 GHz, and across 9–11 GHz S11 stays below -25.3 dB and S23 below -25.2 dB. The isolation resistor therefore matches the input and keeps the outputs isolated across the band, with each output between -3.167 and -3.154 dB.

8. Results – Part B: Optimization

Only the coupling-section length was changed, to 95% and 105% of the initial 4.520 mm. Changing the length moves the frequency at which the section is a quarter wavelength (about 10.5 GHz for −5% and 9.5 GHz for +5%), so the response shifts.

Table 6 Coupler performance at 10 GHz before and after length change

Design	L (mm)	C (dB)	I (dB)	IL (dB)	D (dB)
Initial (L0)	4.520	10.13	37.94	0.587	27.81
−5 % length	4.294	10.15	38.37	0.578	28.22
+5 % length	4.746	10.16	37.51	0.591	27.35

Table 7 Performance over the 9–11 GHz band

Design	Coupling range (dB)	Max |C − 10| (dB)	Mean IL (dB)	Min D (dB)
Initial (L0)	10.13 – 10.23	0.227	0.584	26.8

Design	Coupling range (dB)	Max |C − 10| (dB)	Mean IL (dB)	Min D (dB)
−5 % length	10.12 – 10.33	0.328	0.575	27.4
+5 % length	10.14 – 10.37	0.371	0.587	26.2

Table 8 Band-edge and centre values (C, I, IL, D in dB)

Design	Frequency	C	I	IL	D
Initial (L0)	9 GHz	10.23	38.82	0.577	28.59
Initial (L0)	10 GHz	10.13	37.94	0.587	27.81
Initial (L0)	11 GHz	10.23	37.08	0.577	26.85
−5 % length	9 GHz	10.33	39.22	0.559	28.89
−5 % length	10 GHz	10.15	38.37	0.578	28.22
−5 % length	11 GHz	10.14	37.54	0.579	27.40
+5 % length	9 GHz	10.16	38.42	0.590	28.26
+5 % length	10 GHz	10.16	37.51	0.591	27.35
+5 % length	11 GHz	10.37	36.61	0.569	26.24

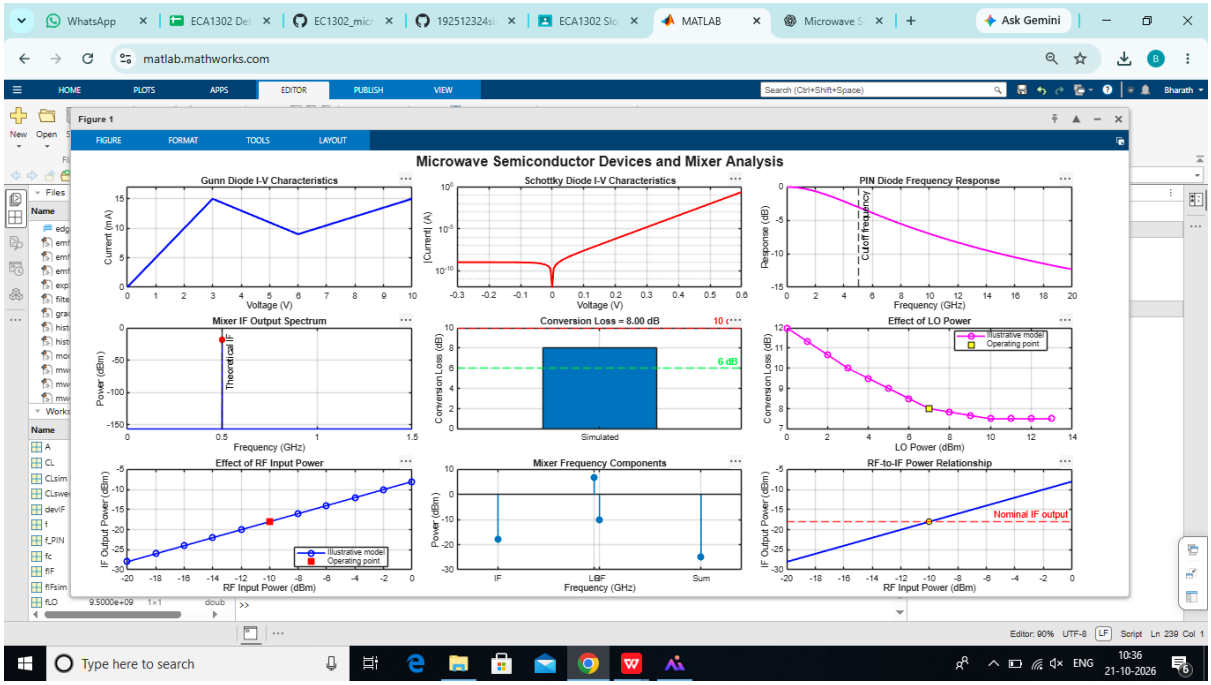


Fig. 6 Coupling, isolation, insertion loss and directivity of the three designs (shaded band: 9–11 GHz).

8.1 Selection of the final design

The initial length gives the smallest worst-case coupling error over 9–11 GHz (0.227 dB, against 0.328 dB for −5% and 0.371 dB for +5%), so it holds coupling closest to 10 dB. The −5% design has marginally lower mean insertion loss (0.575 dB against 0.584 dB), but the difference is only 0.009 dB, while its coupling error is about 45% larger. The +5% design is worse on both counts, with a maximum coupling error of 0.371 dB at 11 GHz. All three designs meet isolation ≥ 25 dB, insertion loss ≤ 1.5 dB and directivity ≥ 15 dB over the whole band.

Final design: the initial quarter-wave length, L = 4.520 mm. It keeps the coupling nearest 10 dB across the band, with an insertion loss within 0.01 dB of the best alternative. The ±5% length change does not improve the response, because the quarter-wave length is already centred on 10 GHz, which is the centre of the 9–11 GHz band.

9. Verification Against Threshold Values

Table 9 Compliance check

Condition	Limit	Result	Status
Ports and sweep	50 Ω, 8–12 GHz	50 Ω, 8–12 GHz	Pass
Coupling deviation	≤ 5% (10 dB)	1.29%	Pass
Isolation	≥ 25 dB	37.9 dB	Pass
Insertion loss	≤ 1.5 dB	0.587 dB	Pass
Directivity	≥ 15 dB	27.8 dB	Pass
Power division deviation	≤ 5% (1:1)	4.77%	Pass
Optimization (9–11 GHz)	Closest 10 dB, lowest IL	Initial length selected	Done

10. Discussion

- Coupler: the coupling of 10.13 dB is 0.13 dB above the target. This comes mainly from line loss, which removes a little power and makes the coupled port slightly weaker than the lossless value.
- Directivity: in an ideal homogeneous coupler the isolated port would receive no power. In the microstrip model the unequal mode velocities leave a small residual signal, so isolation is finite (37.9 dB) but still 12.9 dB above the requirement.
- Bandwidth: coupling changes by less than 0.1 dB across 9–11 GHz because the coupled-line section is a quarter wavelength long at the centre frequency, where the coupling curve is at its maximum and flat.
- Divider: the Wilkinson divider is matched at all three ports (S11 = −45.3 dB) and its outputs are isolated (S23 = −45.3 dB) because the 100 Ω resistor absorbs any reflected power. No power is dissipated in it at 10 GHz when both outputs are equally loaded.
- Optimization: a ±5% length error detunes the quarter-wave condition. The effect on coupling is small, but the worst-case error grows from 0.23 dB to 0.33–0.37 dB, which is why the initial length is retained.
- Limits of the model: this is a circuit-level model. Fringing, discontinuities at the feed junctions and radiation, which a full-wave solver in CST or HFSS includes, are not modelled, so a measured or full-wave result will differ slightly.

11. Conclusion

A 10 GHz, 10 dB coupled-line directional coupler (Z0e = 69.37 Ω, Z0o = 36.04 Ω, L = 4.52 mm) and a two-way Wilkinson divider were designed and simulated from 8 to 12 GHz. The coupler achieved 10.13 dB coupling (1.29% deviation), 37.9 dB isolation, 0.59 dB insertion loss and 27.8 dB directivity. The divider split the power 1:1 at -3.15 dB per output (4.77% deviation) with S11 = -45.3 dB and S23 = -45.3 dB. Changing the coupling length by ±5% did not improve the response, and the initial quarter-wave length was selected as the final design because it keeps the coupling closest to 10 dB over 9–11 GHz. All results are within the specified 5% deviation and the isolation, insertion-loss and directivity limits.

12. Mapping to the Lab Exam Questions

Exam question	Where answered	Section / figure
Q1 Design and simulate 10 dB coupler; compute C, I, IL, D; compare with targets	Tables 2–4, sample calculations	Sec. 5, 7.1; Figs 1, 3, 4
Q2 Design 3 dB divider; ratio, S11, output isolation; compare with −3.01 dB	Tables 2 and 5, sample calculation	Sec. 5, 7.2; Figs 2, 5
Q3 Optimize length by 5%; compare; choose and justify design	Tables 6–8, selection argument	Sec. 8, 8.1; Fig. 6

13. References

1. D. M. Pozar, Microwave Engineering, 4th ed., Wiley, 2012 – chapters on couplers and power dividers.
2. E. J. Wilkinson, "An N-way hybrid power divider," IRE Trans. Microwave Theory Tech., vol. 8, no. 1, pp. 116–118, 1960.
3. Laboratory manual, ECA13 Microwave Engineering – Simulation Experiment 5.